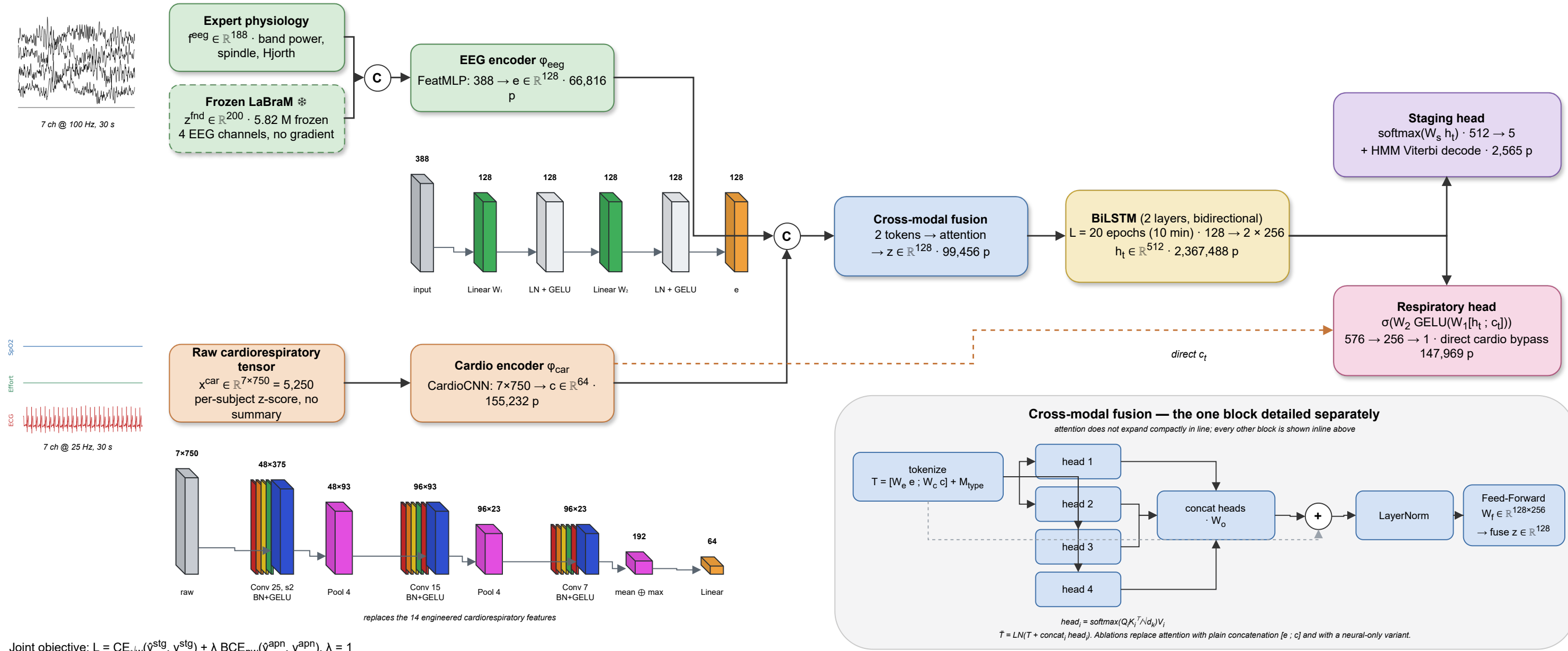


Two-stream multimodal multi-task network · $f_{\theta} : (f_{\text{eeg}}, z^{\text{fnd}}, x^{\text{car}}) \rightarrow (\hat{y}^{\text{stg}}, \hat{y}^{\text{apn}})$



Joint objective: $L = \text{CE}_{\sqrt{w}}(\hat{y}^{\text{stg}}, y^{\text{stg}}) + \lambda \text{BCE}_{\text{pw}}(\hat{y}^{\text{apn}}, y^{\text{apn}})$, $\lambda = 1$

dashed outline = frozen (no gradient) · 2,764,774 trainable + 5,819,936 frozen parameters